Bit Project

Introduction to Seaborn

Module 1

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What is Seaborn?

Learning Objectives:

1. Basics of the Seaborn Package to **plot data**
2. **Value** of Seaborn in the Data Science world.
3. How Seaborn helps us thoroughly conduct **Data Science.**

Some of Seaborn plots

- Factor plots:
  - This allows you to view a distribution of a certain parameter with bins defined by a different parameter.

- Faceted Histograms
  - Seaborn helps with viewing data through histograms containing subsets.

- Pair plots
  - This helps with having joint plots of datasets with larger dimensions.

Seaborn vs Matplotlib

- Matplotlib forms informative plots but they are not aesthetically pleasing.
- With the Seaborn package, we can:
  - **Develop** aesthetic plots.
  - **Deal** with a more convenient Seaborn API.
  - **Visualize** text-based information

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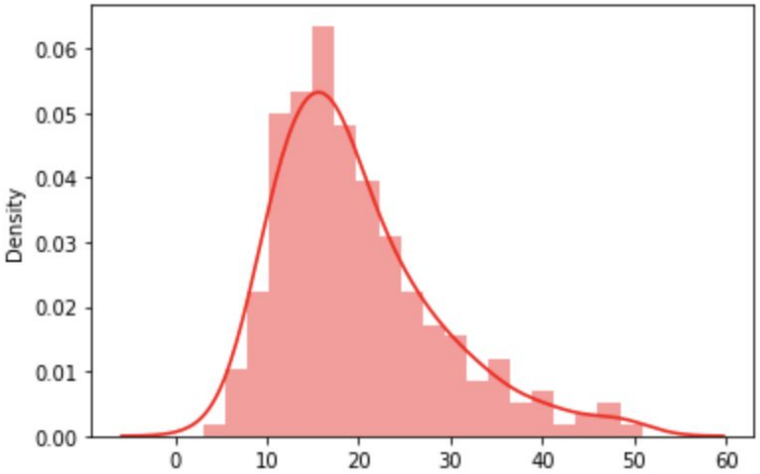
Why, Where, and How we use Seaborn

- Assists in forming data visualizations.
- Can be used for various types of visualizations:
    - Histograms
    - Joint Distributions
    - Bar plots
- Allows for more aesthetically pleasing plots.

Coding Distplot

```
tips = seaborn.load_dataset('tips')
sns.distplot(x=tips['total_bill'],bins=20,color='red')
plt.show()
```

Distplot



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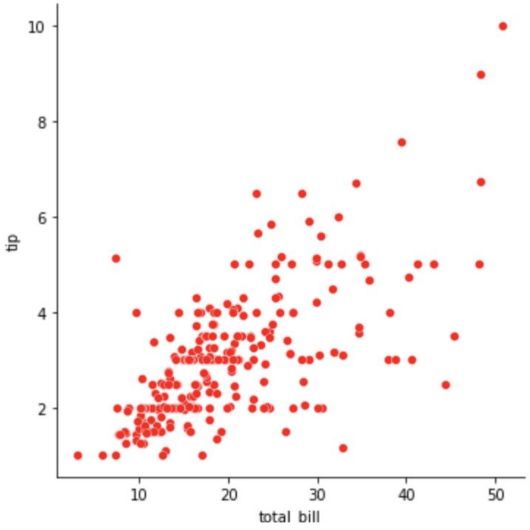
Coding Scatterplot

```
sns.relplot(x="total_bill", y="tip", data=tips, color = 'red')
```

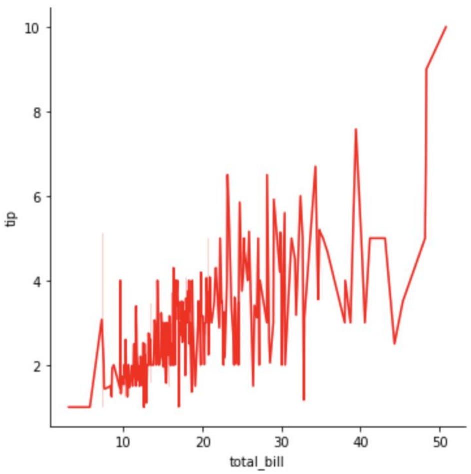
Scatterplot



Coding Lineplot

```
sns.relplot(kind = 'line',x = 'total_bill', y = 'tip',data = tips,color = 'red')
plt.show()
```

Lineplot

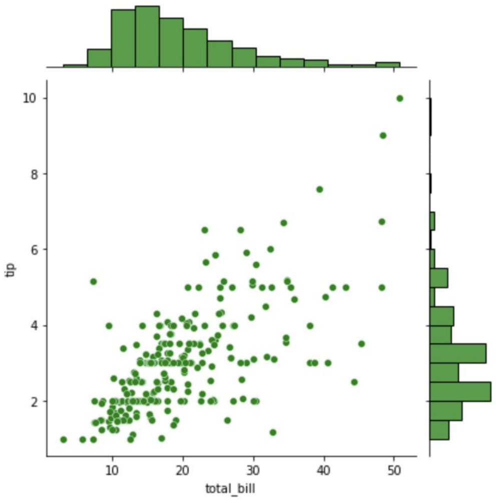


Coding Joint Plot

```python
sns.jointplot(x = "total_bill",
              y = "tip",
              data = tips,
              color = "green")

plt.show()
```

Joint Plot



Coding Boxplot

```
sns.catplot(x="day", y="total_bill", kind="box", data=tips)
```

Boxplot